



# United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

Mid Term Exam Trimester: Fall 2025 Marks: 30 Time: 1 hr 30 mins

Code: CSE 3411 Course Title: System Analysis & Design

**“Any examinee found adopting unfair means will be expelled from the trimester/program as per UIU disciplinary rules.”**

**Answer the following questions:**

1. Consider the following scenario and write down the answers to the questions mentioned as follows:

The **National Library Network (NLN)** has introduced a new digital module to automate two services:

- Users can request to transfer and reserve a book from other branch - **InterBranchBookRequests**
- Users can request to upgrade membership tier - **MembershipTierUpgradeRequests**.

Users can initiate requests directly from their accounts, and the system first performs automatic checks on borrowing limits, outstanding dues, and reservation history. If any issues are detected, the request is halted and the user is notified. All valid requests are then forwarded to the **Branch Operations Unit (BOU)** for review. BOU verifies the availability of resources such as book copies or membership slots. BOU may stop the request and leave an internal remark in the system.

For **InterBranchBookRequests**, the destination branch's BOU checks if a copy is available. If it is then the Logistics Unit schedules the movement through the inter-branch courier system, updating the **Logistics Tracking Table** at each stage. When the book arrives at the destination branch, the reservation is automatically activated for the user.

For **Membership Tier Upgrade Requests**, the BOU consults the **Membership Capacity Table** to confirm quota availability. The Finance Unit calculates the revised membership fee and sends it to the user. Once payment is confirmed, the **Membership Status Table** is updated, and the system activates the user's new privileges, such as extended borrowing limits and access to special digital resources.

After the task-specific processing is complete, the **Central Network Authority (CNA)** performs a final review using the **Network Audit Log Table** to ensure all system actions were executed correctly. Any discrepancies return the request to the relevant unit for correction. Once verified, the CNA approves the request, permanently updating the **User Activity Table**, book records, or membership records, depending on the request type. Automated notifications are then sent to all relevant parties, including the user, BOU, CNA officers, and the Logistics Unit when applicable, completing the fully automated process.

- Draw the **DFD** for the above scenario. [4]
- Draw the **activity diagram** for **InterBranchBookRequests**. [3]
- Draw the **Use case diagram** for the above scenario (show at least one extend/include relationship in the diagram). [3]
- Write down the **descriptive form of one major use case** shown in the use case diagram for the above scenario. [4]

2. a. Considering the above-described scenario was in its initial stage before launch. Briefly explain any **one tool** of information gathering that would be **most beneficial** in gathering the information needed to make this project. Mention the **merits** and **demerits** of the tool you chose. [1+2]
- b. What is the **MoSCoW method**, and why is it used? [2][CO2]
3. Answer the following questions: [CO1]
- a. (i) Mention the steps of the **Software Development Life Cycle (SDLC)**. [1]
- (ii) Explain the key **differences** between the following SDLC steps: [3]
- Design
  - Deployment
  - Maintenance
- b. What is the full form of **ES** and **SaaS**? [1]
4. Suppose you are considering an investment opportunity for a new mobile app startup. The project requires an initial investment of \$10,000 at present, followed by additional investments of \$4,000 at the end of year 1 and \$2,000 more than year one at the end of year 3. In return, the project generates revenues of \$2,000 at the end of year 2, \$3500 at the end of year 3, \$4,500 at the end of year 4, and \$8,000 at the end of year 5. If the annual interest rate for this venture is 9%, use the cash flow method to determine whether you will invest in this project or not. [CO4] [3]
5. Develop and present in the class room on a **partial** SRS document (software requirement specification) as a group wise project assigned in the classroom. The SRS document should cover the following areas regarding your respective project: Introduction, Rationale/purposes of the project, Detail system study (tentatively you should cover benchmark products, research papers, onsite visit, survey, online sources etc.), feasibility analysis (SWOT analysis, cash flow analysis etc.), system design (Use case diagram, use case descriptive form, Data Flow Diagram, activity diagram, swimlane diagram etc.) [CO3] [5]
- ( do not answer this question in the examination)